

Development of Parallel Stance Check System for Ski Videos Using Pose Estimation Model

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Abstract

Parallel stance in ski is pose in which both feet are kept parallel during parallel turn, in which both skis are parallel to each other. Until now, parallel stance quality has been evaluated by human assessment, resulting in issues with subjectivity and reproducibility. In this study, I developed system for checking parallel stance quality from ski videos. The proposed method was able to track target by playback direction adjustment based on filming method and using ROI. After comparing 2D or 3D pose estimation model, I determined to use 2D pose estimation model. The judgment is made by focusing on angle between both shins from estimated pose. The angle threshold was determined from histogram that tallying number of occurrences of parallel and not parallel for each angle. Incorrect pose estimation largely be cleaned by using median filter. Limitations of proposed method include possibility of tracking being interrupted if target is located too far away. 2D pose estimation model is weak against side on pose estimation. Since judgement criteria of parallel or not parallel are subjective, system's judgment is also subjective. If incorrect pose estimations are made repeatedly, they cannot be cleaned, resulting in an incorrect judgment. In conclusion, proposed method can track target, and system was almost possible to make appropriate judgments based on form of skier in many frames. In the future, I hope to develop more general-purpose system by introducing 3D pose estimation model, relaxing of ski videos requirements and improving judgment criteria.

1 Introduction

1.1 Background and Problems

In ski, parallel turn is fundamental and important turn technique performed while maintaining both skis parallel. Parallel Stance (PS) is pose in which both feet are kept parallel during parallel turn [1]. PS quality has been evaluated by human assessment which tends to rely on personal experience and subjectively, resulting in inconsistent assessments even for same ski performance. Furthermore, it is difficult for skiers to objectively assess their form

from their ski videos, making it hard to quantitatively grasp the extent to which PS is maintained at each stage. Like this, checking PS face challenges in terms of reproducibility and objectivity.

1.2 Recently Development of AI and Related Research

Recent advances in deep learning have significantly improved human pose estimation from images and videos. In addition, research related ski using pose estimation model is also being conducted to predict the positions of skis and poles from estimated pose [2]. Despite these efforts, no research has been conducted aimed at automated to check PS.

1.3 Goal

By using pose estimation technology, it may be possible to automatically check PS and objectively evaluate form. In this study, I developed system for checking PS quality from ski videos. The purpose of this study is to check PS in objective and reproducible manner and can be used practically in technical instruction by instructors and in the self-practice of skiers themselves. By objectively checking PS, which previously relied on experience and subjectivity, I aim to efficiently support skiers in understanding their technique and improving it.

2 Design

2.1 System Overview

First, I created simple system flowchart as shown in Fig. 1.

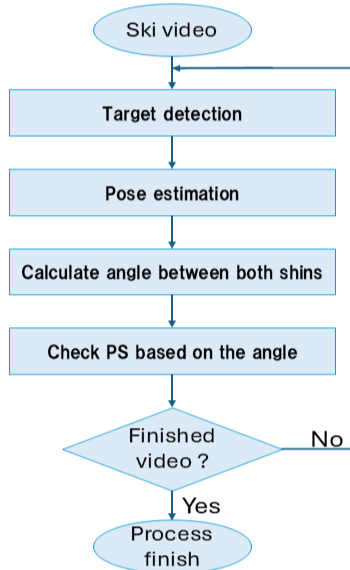


Fig. 1. System overview

2.2 Problem Definition

When developing PS check system, the following issues are likely to arise:

1. Tracking may fail due to other skiers or small target in target detection process.
2. Which pose estimation model to use such as 2D and 3D in pose estimation process.
3. How to check PS based on angle between both shins (angle) in check PS based on the angle process.
4. How to deal with incorrect pose estimation in check PS based on the angle process.

Each issue is treated through three stages:

1. Solution to issue 1 is explained in 2.4.1, its effectiveness is explained in 3.1, and discussion of the results and limitations is given in 5.1.1.
2. Solution to 2 is explained in 2.4.2, determinations are presented in 3.2, and discussion of the results and limitations is given in 5.1.2.

3. Solution to 3 is explained in 2.4.3, determinations are presented in 3.3, and discussion of the results and limitations is given in 5.1.3.
4. Solution to issue 4 is explained in 2.4.4, its effectiveness is explained in 3.4, and discussion of the results and limitations is given in 5.1.4.

I would like to solve these issues for develop system that can easily perform to check PS such as parallel or not parallel on ski videos.

2.3 Ski Videos Requirements for Reliable Pose Estimation

System requires the following ski videos for reliable pose estimation.

- Cameraman at bottom of course films target at top course ski as shown in Fig. 2 (ski video filmed from below).



Fig. 2. Screenshot of ski video filmed from below

- Target and cameraman at top of course, and cameraman films target ski. This also includes ski videos where cameraman follows behind target as shown in Fig. 3 (ski video filmed from behind).



Fig. 3. Screenshot of ski video filmed from behind

These viewpoints make pose estimation easier than side, regardless of whether 2D or 3D pose estimation model.

2.4 Solutions

2.4.1 Playback Direction Adjustment Based on Filming Method and Target Detection Using ROI

For ski video filmed from below, target appears larger in around final frames, so target is tracked by reverse playback. For ski video filmed from behind, it is no change, so target is tracked by playback. Next, I considered YOLO's yolo12n.pt [3][4] for object detection model, which is very accurate and fast. Also, taking advantage of the fact that target hardly moves between frames, Region of Interest (ROI) is designed by multiplying diagonal of bounding box of YOLO detected in previous frame by 1.5, and this is executed every frame to track the target. Since other skiers may appear, user is required to manually input about location of target for last frame of ski video filmed from below and first frame of ski video filmed from behind.

2.4.2 Comparing 2D and 3D Model to Determine Which Model to Use

I compare yolo11x-pose.pt of YOLO [5], which can estimate 2D pose, and Blaze Pose's pose_landmarker_full.task [6][7], which can estimate 3D pose. After comparing, I determine which pose estimation model to use.

2.4.3 Determine Angle Thresholds

To check PS based on angle, angle threshold that angles below this number are parallel and angles above this number are not parallel is determined in the following way.

Preparation: I use following eight ski videos as input for determining angle thresholds.

- Two ski videos filmed from below of not parallel skier
- Two ski videos filmed from behind of not parallel skier
- Two ski videos filmed from below of parallel skier
- Two ski videos filmed from behind of parallel skier

1. Obtain all frames (494 frames) from above ski videos.
2. Subjectively judge whether it is parallel or not parallel and label it.
3. Estimate pose of all frames using pose estimation model and calculate angle.
4. To visualize number of occurrences of parallel and not parallel for each angle by creating histogram.
5. Determine angle thresholds from histogram.

2.4.4 Cleaning Incorrect Pose Estimation by Median Filter

Pose estimation model may estimate incorrect pose. In result, system gives incorrect judgements so median filter is applied to judgements. Window size of median filter is two. If judgments of previous and next frames are same, and judgment of only center frame is different, the judgement is considered to be the same as previous and next frames.

3 Results

In this chapter, I show how effective solutions are shown in 2.4 and also determine which pose estimation model to use and angle thresholds for checking PS.

3.1 Playback Direction Adjustment Based on Filming Method and Target Detection Using ROI

This shows how effective proposed method described in 2.4.1 was. Fig. 4 shows comparison of target detection test for ski video filmed from below by playback & no ROI (Original) and reverse playback & ROI (Propose method). Confidence score 0 indicates a case where target detection failed. In Original, confidence score fluctuates between around 0.6 and 0.8, and target detection fails around 16 to 18 seconds of ski video. On the other hand, Proposed method maintains confidence score above 0.8, also indicating that target detection is successful throughout. Also, from Table. 1, It can confirm that Proposed method has higher average confidence score and target detection success rate than Original. Based on these results, I determined to use Proposed method for target detection.

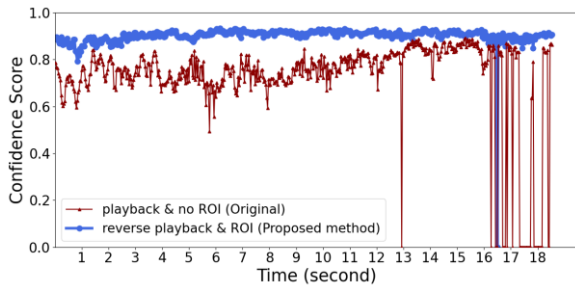


Fig. 4. Comparison of target detection test by Original and Proposed method

Table. 1. Average confidence score and target detection success rate of Proposed method and Original

	Average confidence score	Target detection success rate
Proposed method	0.904	99.82 %
Original	0.77	93.53 %

3.2 Comparing 2D and 3D Model to Determine Which Model to Use

To determine which pose estimation model to use, **Fig. 5** compared accuracy of pose estimation between `pose_landmarker_full.task` and `yolo11x-pose.pt`. Since `pose_landmarker_full.task` does not display confidence score like `yolo11x-pose.pt`, I averaged all joint point scores and displayed them in the upper left. From **Fig. 5**, it can be visually confirmed that `pose_landmarker_full.task` is unable to estimate correct joint positions and skeleton, especially in foot area, but `yolo11x-pose.pt` is able to estimate correct joint positions and skeleton overall. These results, I determined to use `yolo11x-pose.pt` as pose estimation model in this study.

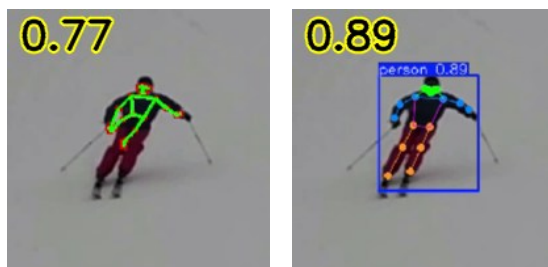


Fig. 5. Comparison of accuracy of pose estimation between `pose_landmarker_full.task` (left) and `yolo11x-pose.pt` (right)

3.3 Determine Angle Thresholds

To check PS based on angle, I created a histogram of **Fig. 6** as described in 2.4.3 and determine angle thresholds from it. Parallel ratio refers to how many parallels are included in one section and is plotted as blue line graph. Green line is 50% of parallel ratio. Less than 13° is almost parallel. Around 14° to 15° , parallel and not parallel balance is half and half. More than 16° is perfectly not parallel. These results, angle threshold is:

angle $< 14^\circ \Rightarrow$ parallel

$14^\circ \leq \text{angle} < 16^\circ \Rightarrow$ controversial form

$16^\circ \leq \text{angle} \Rightarrow$ not parallel

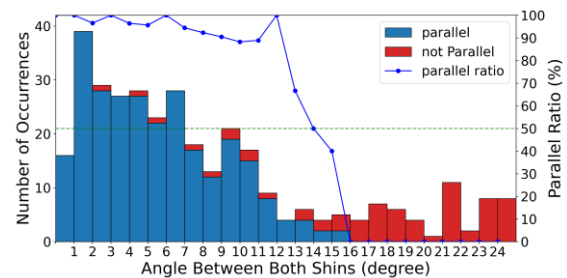


Fig. 6. Histogram of parallel & not parallel distribution for each angle and line graph of parallel ratio

3.4 Cleaning Incorrect Pose Estimation by Median Filter

To confirm effectiveness of median filter, I compared judgements of original and median filtered. **Fig. 7** shows comparison of judgements of original and median filtered for not parallel skier. Even though skier was not parallel, there were moments during turn when skier was filmed from side, causing `yolo11x-pose.pt` to mistakenly estimate pose as parallel. Therefore, before median filtering, judgements is parallel around 1.1, 1.2, 1.5, and 1.9 seconds. If I leave this, the system would mistakenly judge parallel, even though actually skier was not parallel. After median filtering, incorrect judgements was cleaned. But repeatedly incorrect pose estimation around 1.2 second could not be cleaned. However, since it is effective for single incorrect pose estimation, I will use this method.

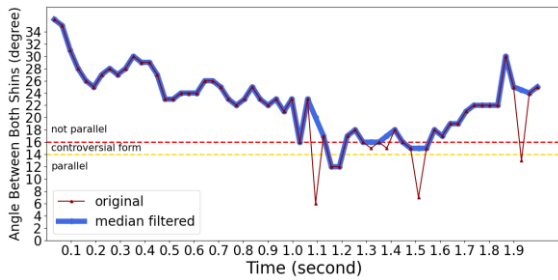


Fig. 7. Comparison of judgements of original and median filtered for not parallel skier

4 Developed PS Check System

4.1 Front-end

On front-end, user must enter some items to check PS, as shown in Fig. 8.

1. Upload a ski video.
2. Trim ski video if necessary.
3. Select filming method. If filming method is from below, target is tracked by reverse playback.
4. Input initial ROI based on filming method because other skiers may appear.
5. Press start button.

If all required item is not entered, error message is displayed and checking PS will not start.



Fig. 8. UI of PS check system

4.2 Back-end

Back-end program receives a ski video and initial ROI from front-end and processes the flowchart of Fig. 9. To inform users of result of checking PS, frame is colored based on judgement.

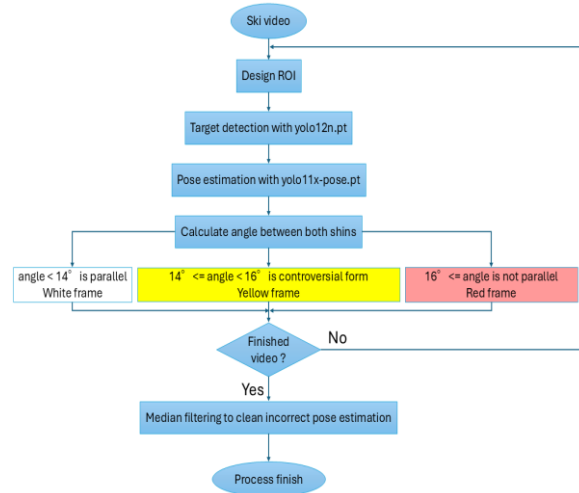


Fig. 9. Flowchart of PS check system

After processing Fig. 9, back-end returns colored ski video and parallel ratio that are results of checking PS like Fig. 10.



Fig. 10. Screenshot of PS check result page

5 Discussion

5.1 Discussion of Results and Limitations

5.1.1 Playback Direction Adjustment Based on Filming Method and Target Detection Using ROI

It is believed that yolo12n.pt is able to recognize target more easily by proposed method, resulting in an increase in confidence score and target detection success rate. However, if target is located too far away, yolo12n.pt can't recognize target using ROI, and tracking is interrupted.

5.1.2 Comparing 2D and 3D Model to Determine Which Model to Use

yolo11x-pose.pt is excellent not only in pose estimation but also in object detection, which is why it was able to recognize target even if they were located a little far away. On the other hand, pose_landmarker_full.task is lightweight for mobile devices and is designed to assume that input is from people who are very close to cameraman. Therefore, if target is located even slightly far away, it cannot be accurately recognized, and pose estimation inevitably fails. Although developed PS check system in this study uses yolo11x-pose.pt, issue of 2D pose estimation remains. For example, even if skier is not parallel when viewed from front, it may appear parallel when viewed from side. Also, if target is located too far away, image quality becomes coarse, making accurate pose estimation difficult. In other words, there is possibility that the system may make incorrect judgements.

5.1.3 Determine Angle Thresholds

I analyzed features from the histogram in Fig. 6 and were able to determine angle thresholds required for checking PS. While angles between 2° and 13° were treated as parallel in this study but there are some not parallel histograms in the range. Visual inspection of relevant frames confirmed that yolo11x-pose.pt had mistakenly estimated the pose as parallel, even if actual form was not parallel. Reason was that there were moments during turn when skier was filmed from side. This is disadvantage of 2D pose estimation as mentioned above. In determining angle thresholds, I tallied number of occurrences of parallel and not parallel for each angle but labeling of form of skier as parallel or not parallel was subjective. As result, there is possibility that boundary between parallel and not parallel will vary slightly from individual to

individual. This inevitably means that there may be some mismatch between judgments of PS check system and individual judgments.

5.1.4 Cleaning Incorrect Pose Estimation by Median Filter

If correct pose estimation had been performed before and after, it is confirmed that incorrect pose estimation that occurs in the middle frame could have been cleaned by median filter. However, if incorrect pose estimation occurs repeatedly, it cannot be cleaned by median filter.

5.2 Practical Conditions

From the above, it can be said that this is a practical system that can accurately checking PS if target is skiing not far from cameraman and ski videos is recorded using specified filming method.

5.3 How Can It be Used for Teaching and Self-Practice

This study did not involve having actual users try it out to verify its effectiveness. However, if this system is generally released, it will automatically check PS and visualize at which point during performance (first half/second half, right/left turns, etc.) PS is not being maintained. This will allow instructors to objectively share points to correct for lessons and enable skiers to quickly identify their issues and bad habits during self-practice.

6 Conclusion

6.1 What I Did in This Study

In this study, I developed system for checking PS quality from ski videos so that instructors and skiers can easily check PS.

6.2 What was Achieved

The proposed method used playback direction adjustment based on filming method and ROI to improve target detection success rate and confidence score so that yolo12n.pt could efficiently recognize target from ski videos. In addition, by focusing on the angle between both shins from pose estimated by yolo11x-pose.pt, confirmed that it was almost possible to make appropriate judgments based on form of skier in many frames.

6.3 Future Challenges

Because 2D pose estimation is used in this system, there are issues of incorrect judgments when input ski video had some scene that skier is filmed from side or when target is located too far away. Furthermore, if incorrect pose estimations occur repeatedly, cleaning by median filter is not possible, which may result in incorrect judgments. Future challenges include introducing new 3D pose estimation model, relaxing ski videos requirements and improving judgments criteria, which are expected to lead to development of more versatile PS check system.

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